Experiment 06: To develop an android application that creates an alert upon receiving a message

PART A

A.1 Aim: To develop an android application that creates an alert upon receiving a message.

A.2 Objectives: To introduce students with various tools like Android Studio, NS2, Wireshark, Cisco packet tracer, WAP supported browser etc.

A.3 Outcomes:

After successful completion of this experiment students will be able to develop an android application that creates an alert upon receiving a message

A.4Theory:

SOFTWARE:

- Android Studio
- The Android SDK (Starter Package)
- Gradle
- Java Development Kit (JDK) 5

DESCRIPTION:

- Open android studio and select new android project . • Give project name and select next
- Choose the android version.
- Enter the package name. package name must be two word separated by comma and
click finish
- Go to package explorer in the left hand side and select our project. • Go to res folder and select layout. Double click the main.xml file
- Now you can see the Graphics layout window.

INPUT:

MAINACTIVITY.JAVA

```
package com.alert;

import androidx.appcompat.app.AppCompatActivity; import
android.os.Bundle; import android.app.Notification;
import android.app.NotificationManager; import
   android.app.PendingIntent; import android.content.Intent;
import android.view.View; import android.widget.Button; import
android.widget.EditText;

public class MainActivity extends AppCompatActivity{
Button notify; EditText e;

@Override
protected void onCreate(Bundle savedInstanceState) {
   super.onCreate(savedInstanceState); setContentView(R.layout.activity_main);
notify= (Button) findViewById(R.id.button); e= (EditText)
findViewById(R.id.editText);

notify.setOnClickListener(new View.OnClickListener() {
@Override public void onClick(View v) {
Intent intent = new Intent(MainActivity.this, SecondActivity.class);
PendingIntent pending = PendingIntent.getActivity(MainActivity.this, 0, intent, 0);
Notification noti = new
Notification.Builder(MainActivity.this).setContentTitle("New
Message").setContentText(e.getText().toString()).setSmallIcon(R.mipmap.ic_l
auncher).setContentIntent(pending).build();
NotificationManager manager = (NotificationManager)
getSystemService(NOTIFICATION_SERVICE);
noti.flags |= Notification.FLAG_AUTO_CANCEL; manager.notify(0, noti);
}
});
}
}
```

ACTIVITY_MAIN.XML

```xml
<?xml version="1.0" encoding="utf-8"?>
android:layout_margin="10dp" android:orientation="vertical">

<TextView
android:layout_width="wrap_content"          android:layout_height="wrap_content"
android:text="Message" android:textSize="30sp" />

<EditText
android:id="@+id/editText" android:layout_width="match_parent"
android:layout_height="wrap_content"                              android:singleLine="true"
android:textSize="30sp" />

<Button
android:id="@+id/button"                              android:layout_width="wrap_content"
android:layout_height="wrap_content" android:layout_margin="30dp"
android:layout_gravity="center"   android:text="Notify"   android:textSize="30sp"/>
</LinearLayout>
```

ANDROIDMANIFEST.XML

```xml
<?xml version="1.0" encoding="utf-8"?>
<manifest              xmlns:android="http://schemas.android.com/apk/res/android"
package="com.alert">

<application
android:allowBackup="true"   android:icon="@mipmap/ic_launcher"
android:label="@string/app_name"
   android:roundIcon="@mipmap/ic_launcher_round" android:supportsRtl="true"
android:theme="@style/Theme.Alert">
<activity
android:name=".SecondActivity" android:exported="true" />
<activity
android:name=".MainActivity" android:exported="true">
<intent-filter>
<action android:name="android.intent.action.MAIN" />

<category android:name="android.intent.category.LAUNCHER"/>
</intent-filter>
</activity>
</application>
```

CREATING A NEW ACTIVITY

Create a new activity named “SecondActivity”

File -> New -> Activity -> Empty Activity

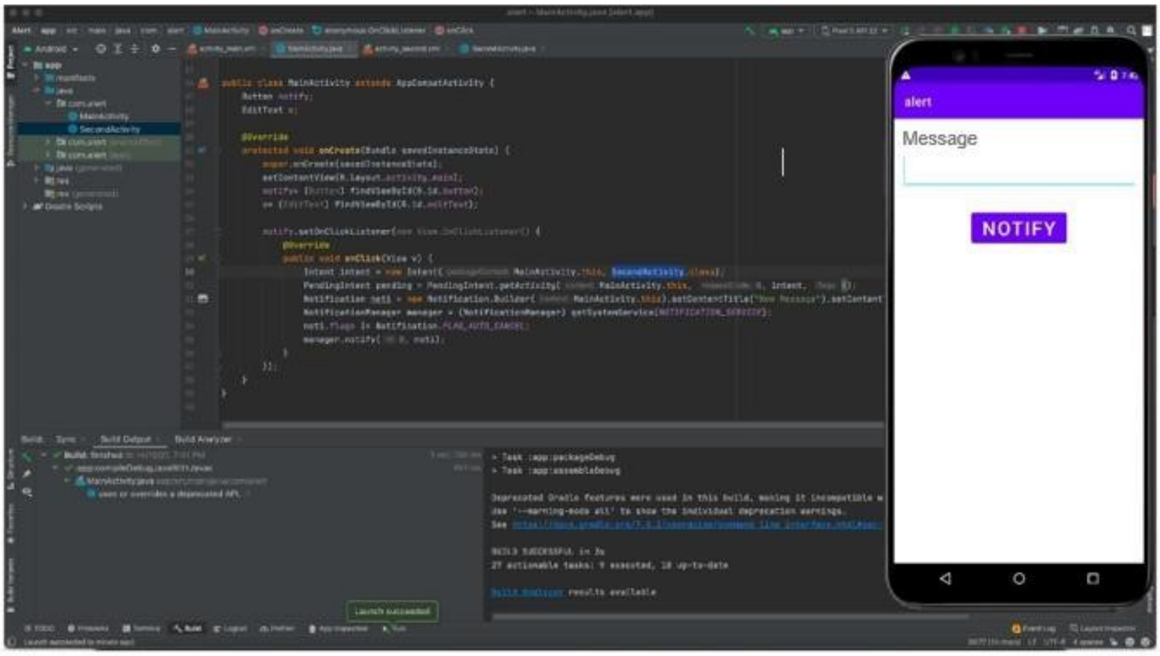
SECONDACTIVITY.JAVA

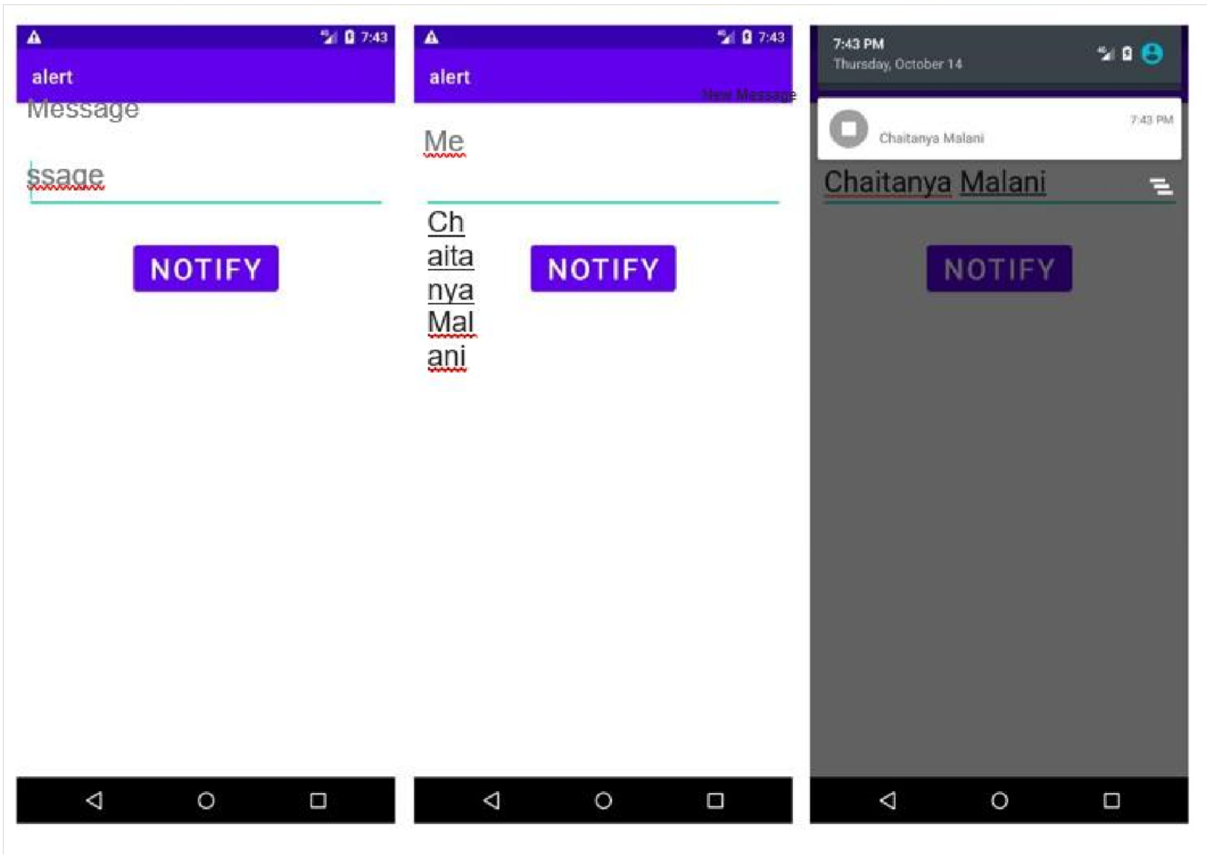
```
package com.alert;

import androidx.appcompat.app.AppCompatActivity; import
android.os.Bundle; public class SecondActivity extends
AppCompatActivity{ @Override
protected void onCreate(Bundle savedInstanceState) {
   super.onCreate(savedInstanceState); setContentView(R.layout.activity_second);
}
}
```

ACTIVITY_SECOND.XML

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayoutxmlns:android="http://
schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
   android:layout_width="match_parent" android:layout_height="match_parent"
tools:context=".SecondActivity">

</androidx.constraintlayout.widget.ConstraintLayout>
```

OUTPUT:





PART B

(PART B: TO BE COMPLETED BY STUDENTS)

(Students must submit the soft copy as per following segments within two hours of the practical. The soft copy must be uploaded on the Blackboard or emailed to the concerned lab in charge faculties at the end of the practical in case the there is no Black board access available)

Roll. No. C54	Name: Quazi Md Moinuddin
Class : TE-C	Batch: C3
Date of Experiment: 25-2-25	Date of Submission: 4-2-25
Grade:	

B.1 Software Code written by student/steps:

(Paste your Code script related to your case study completed during the 2 hours of practical in the lab here)

Activity_main.xml :

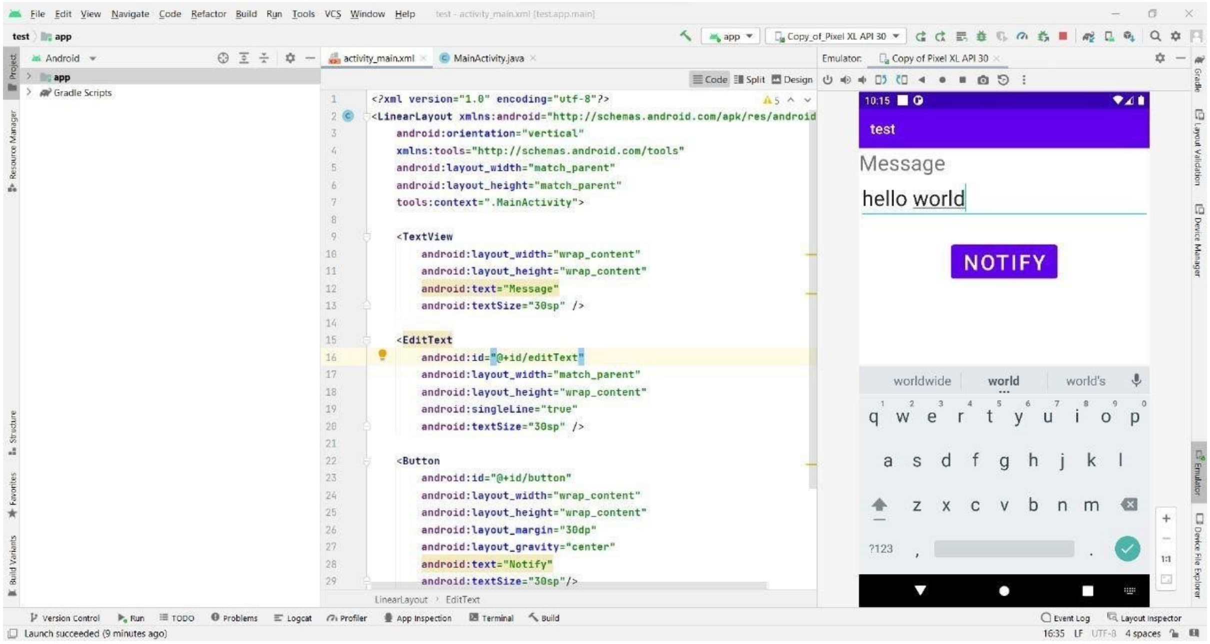
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
android:orientation="vertical"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent" android:layout_height="match_parent"
tools:context=".MainActivity">

<TextView android:layout_width="wrap_content" android:layout_height="wrap_content"
android:text="Message"
android:textSize="30sp" />

<EditText android:id="@+id/editText" android:layout_width="match_parent"
android:layout_height="wrap_content" android:singleLine="true" android:textSize="30sp"
/>

<Button android:id="@+id/button"
android:layout_width="wrap_content" android:layout_height="wrap_content"
android:layout_margin="30dp" android:layout_gravity="center" android:text="Notify"
android:textSize="30sp"/>

</LinearLayout>
```

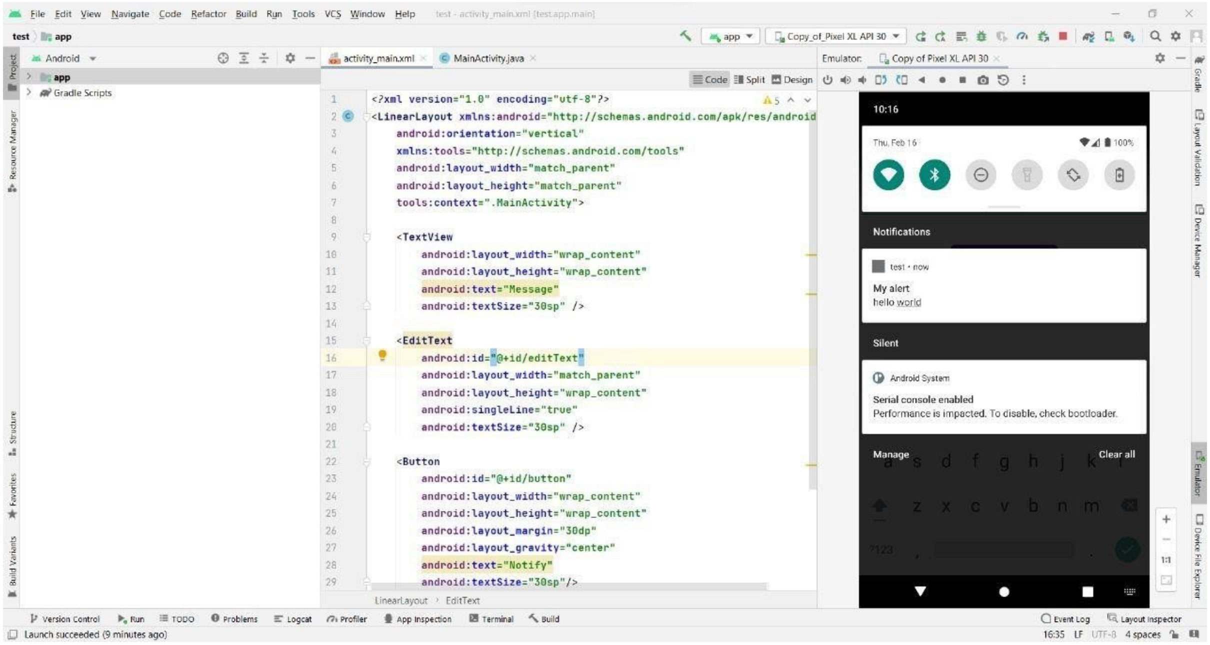
MainActivity.java

```
package com.vikas.test;

import androidx.appcompat.app.AppCompatActivity; import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;

import android.app.NotificationChannel; import android.app.NotificationManager;
import android.os.Build; import android.os.Bundle; import android.view.View; import
android.widget.Button;  import android.widget.EditText;


public class MainActivity extends AppCompatActivity {
Button notify;
EditText e;

@Override protected void onCreate(Bundle savedInstanceState)
{ super.onCreate(savedInstanceState); setContentView(R.layout.activity_main);
notify=findViewById(R.id.button); e =findViewById(R.id.editText);

if(Build.VERSION.SDK_INT>=Build.VERSION_CODES.O){
NotificationChannel channel = new NotificationChannel("My notification","My notification",
NotificationManager.IMPORTANCE_DEFAULT);
NotificationManager manager = getSystemService(NotificationManager.class);
manager.createNotificationChannel(channel); }

notify.setOnClickListener(new View.OnClickListener() {
@Override
public void onClick(View v) {
NotificationCompat.Builder builder = new
NotificationCompat.Builder(MainActivity.this,"My notification"); builder.setContentTitle("My alert");
builder.setContentText(e.getText());  builder.setSmallIcon(R.drawable.ic_launcher_background);
builder.setAutoCancel(true);

NotificationManagerCompat managerCompat =
NotificationManagerCompat.from(MainActivity.this); managerCompat.notify(23,builder.build()); }
});


}
}
```

B.2 Input and Output:

(Paste your output that you are getting after running app in from of screenshots.)





B.3 Observations and learning:

(Students are expected to comment on the output obtained with clear observations and learning for each task/ sub part assigned)

The experiment involves developing an Android application that creates an alert upon receiving a message. Through the provided code, students learn to utilize Android Studio, implement notification functionalities, and handle user input effectively.

B.4 Conclusion:

(Students must write the conclusion as per the attainment of individual outcome listed above and learning/observation noted in section B.3)

Students gain practical experience in Android app development, understanding how to create notifications triggered by user actions. This experiment enhances their skills in utilizing Android SDK tools and Java programming for mobile application development.

B.5 Question of Curiosity

(To be answered by student based on the practical performed and learning/observations)

1) Explain different steps required to build up this alert project?

Ans :

The steps to build the alert project in the provided experiment are as follows:

1. Set up the development environment with Android Studio and necessary SDKs.
2. Create a new Android project in Android Studio.
3. Design the user interface using XML layout files.
4. Implement the logic for receiving messages and generating alerts in the MainActivity.java file.
5. Configure notification channels and manager for displaying alerts.
6. Test the application on an emulator or physical Android device.
7. Debug and refine the code as necessary to ensure proper functionality.